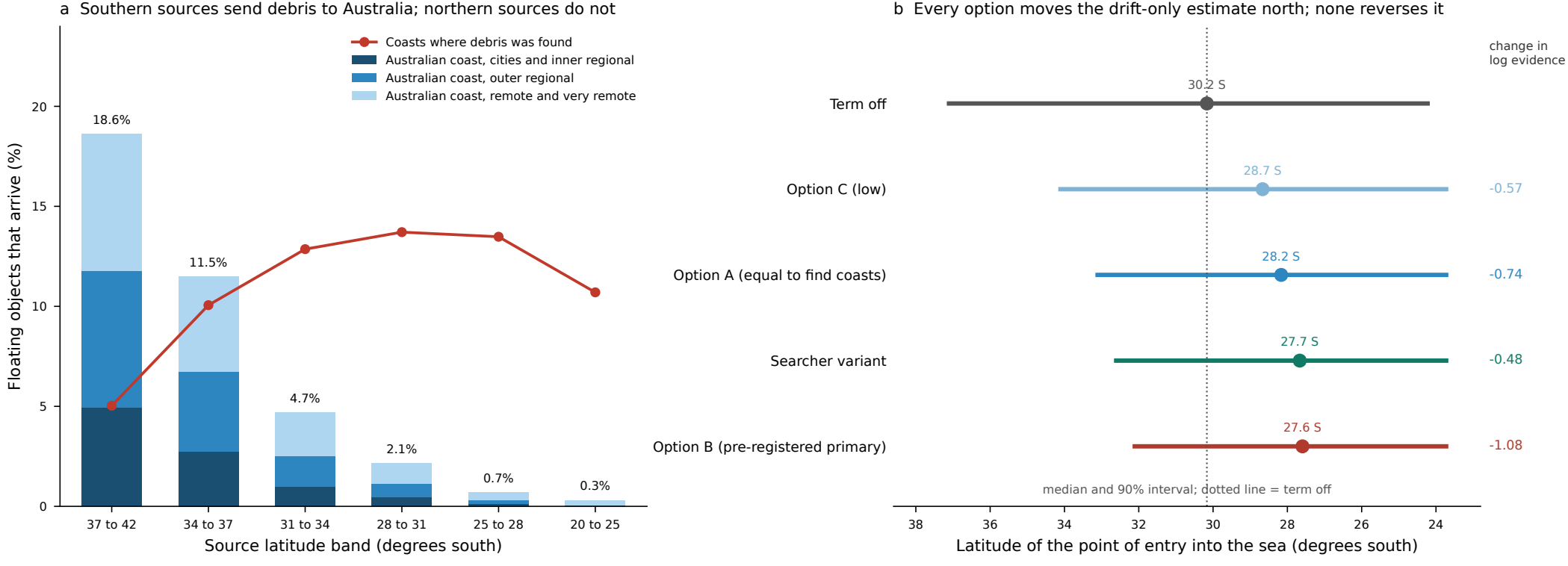
Australian non-recovery term: the northward movement survives every pre-registered weighting



Plain-language note. A source node is a place where the aircraft could have entered the sea. For each node the model releases floating objects and follows them with ocean currents and wind. Australia has no confirmed debris, so nodes that send many objects to Australian coasts lose support. How much support they lose depends on how likely a find there was to be seen and reported. That likelihood is set by the remoteness of the coast. The chart tests four pre-registered settings of it, plus one setting that switches the Australian coasts off. The estimate moves north in all of them.

Technical note. Debris drift module, Australian non-recovery term (gap G26), pre-registered options of 12 Oct 2026 (B primary: city and inner regional ln 1, outer regional ln 0.3, remote ln 0.05; A: find-coast levels ln 0.1/0.5/1 by period; C: ln 0.3/0.1/0.02; searcher: A plus ln 3 on southern Mozambique and north-east Madagascar in Mar-Jun 2016; sigma 1, 256 common draws). Source: completed interim pilot of 12-13 Oct 2026 (GLORYS12 + ERA5, 175 of 178 nodes scored in all options, 1e5 particles per node, window 8 Mar 2014 - 30 Jun 2016, beaching on entry to the model land mask, no refloat); no new transport was run. Panel a: mean over the three object classes. Panel b: run A GLORYS12 node likelihoods with the per-node change carried by inverse-distance weighting from the pilot nodes, uniform prior over 367 recoverable run A nodes (the full 710-node surface is not on disk), cos(latitude) area weight. Monte Carlo noise of the mean-latitude shift 0.35-0.49 deg (split-half). REACH-LIMITED: domain ends at 120 E, so Australian south coast and Tasmania are unreachable and 42% of particles from 37-42 S leave the domain; the southern penalty is probably understated. Impact prior not applied; not composed with other evidence; no Bayes factor. Interim, not evidence: run I is not converged and the end-of-flight uncontrolled arm is not calibrated.